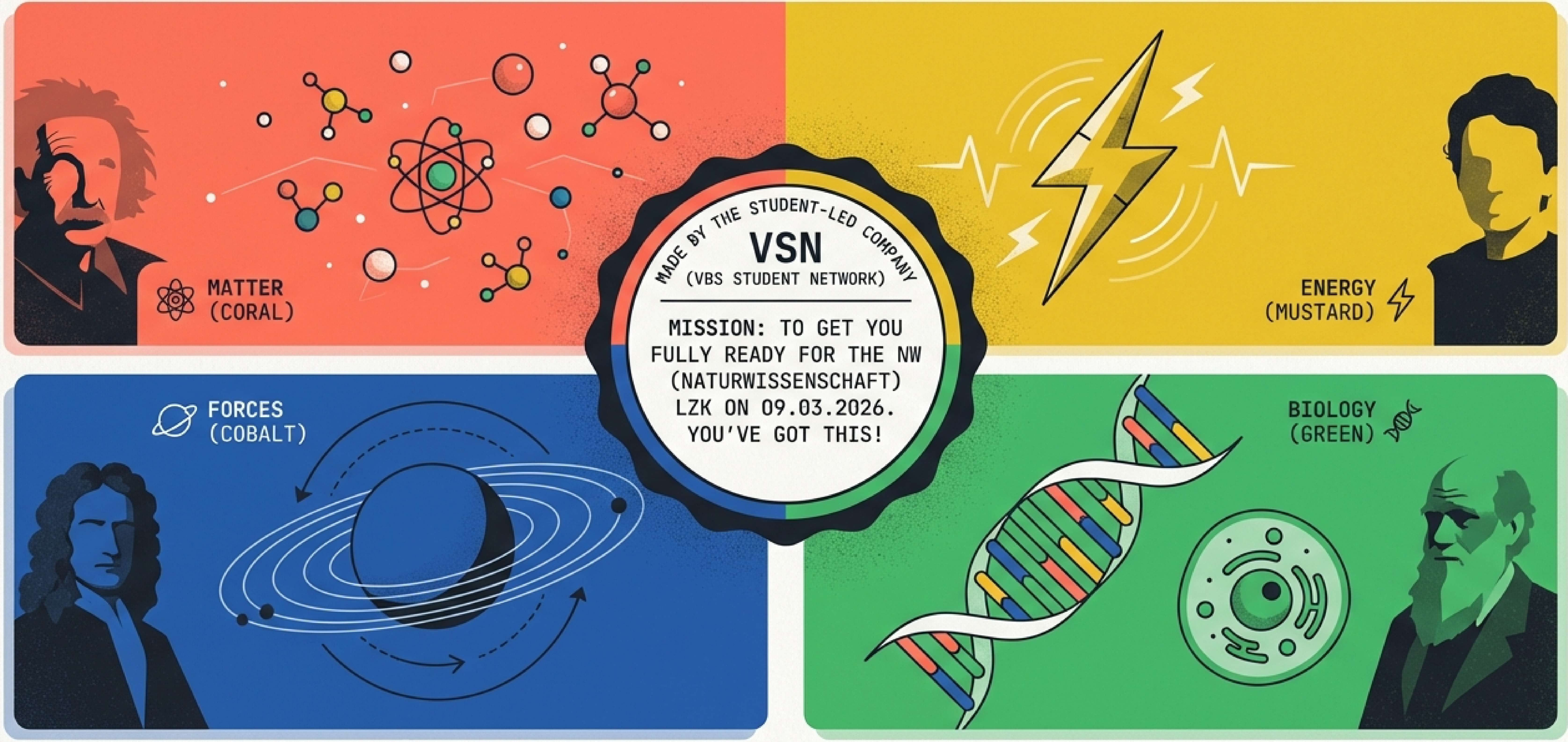
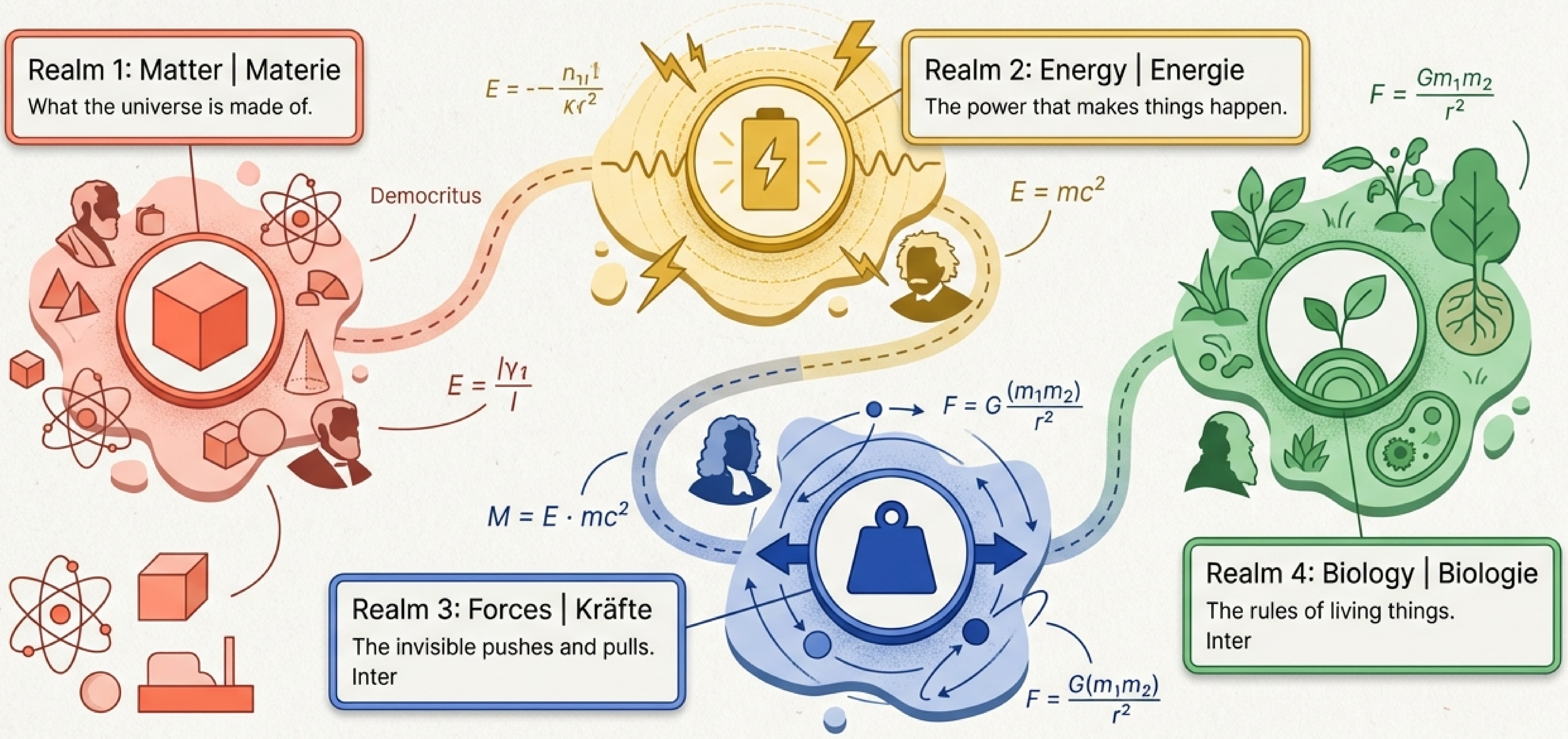
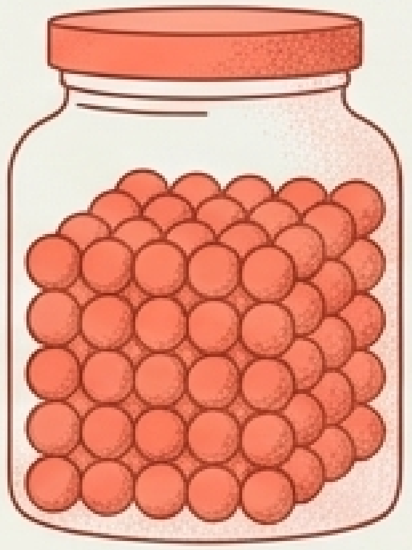
THE ULTIMATE SCIENCE SURVIVAL GUIDE



Your Map of the Universe



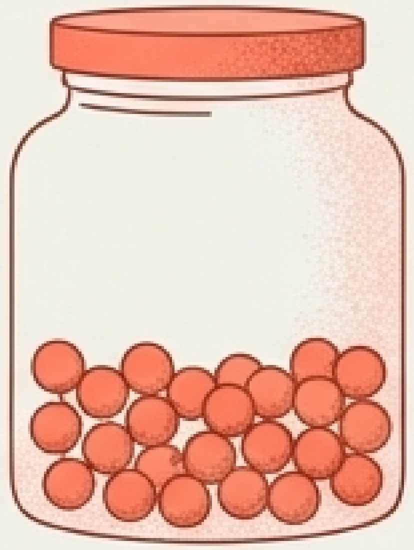
Realm 1: States of Matter | Aggregatzustände



Solid | Fest

Particles are locked in a tight grid. Constant shape, constant volume.

(Example: Ice, Metal).



Liquid | Flüssig

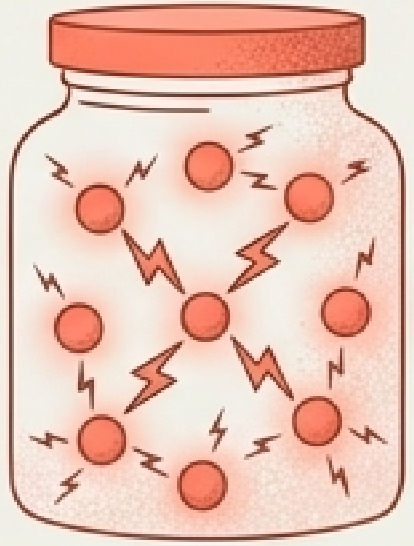
Particles are close but slide around. Variable shape, constant volume.

(Example: Water).

Gas | Gasförmig

Particles fly everywhere, taking up 1000x more space! Variable shape and volume.

(Example: Air).

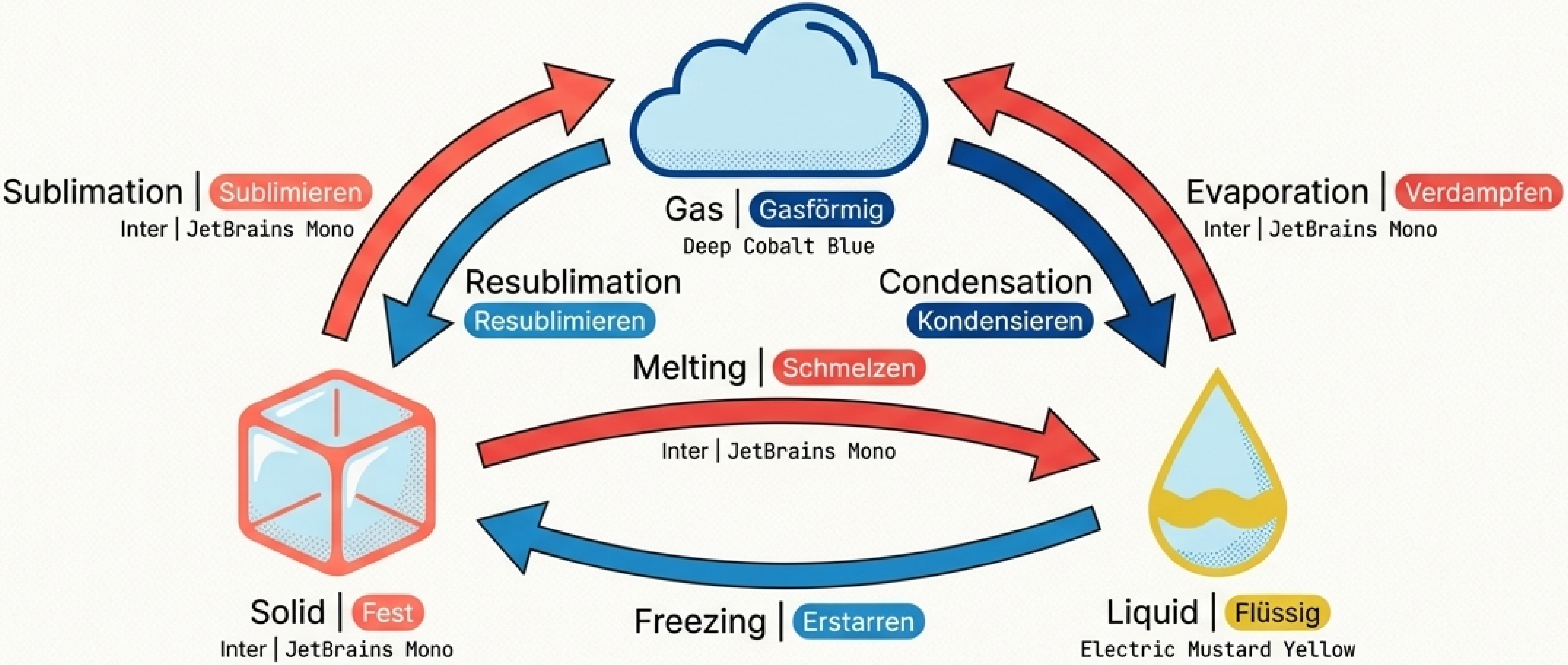


Plasma | Plasmaförmig

Supercharged, glowing particles. Makes up 98% of the visible universe!

(Example: Neon lights, the Sun).

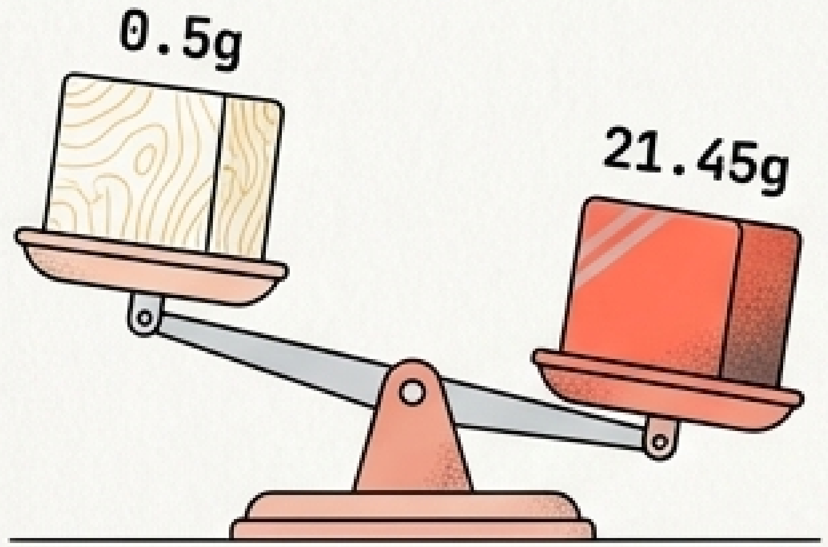
Phase Changes | Übergänge



Fun Fact: `Helium` melts at an insane `-272°C`, while `Tungsten (Wolfram)` melts at a blazing `3422°C`!

Properties of Matter | Stoffeigenschaften

$$\rho = \frac{m}{V}$$

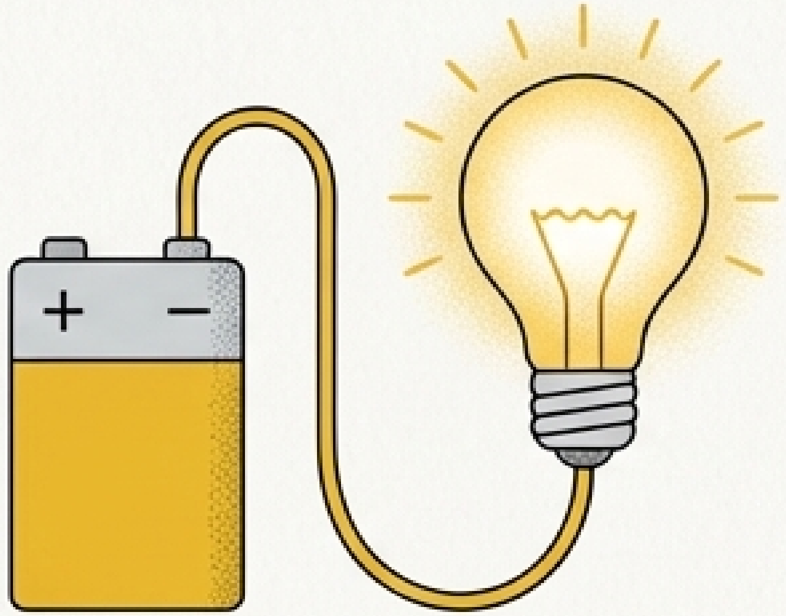


Density | Dichte

Density = Mass divided by Volume.

Measured in kg/m³.

Conductivity | Leitfähigkeit



Does it let electricity flow?

Metals are great conductors. Rubber and Glass are insulators (Nichtleiter).

Measured in Siemens (S).

Viscosity | Viskosität

How "thick" a liquid is.

Honey = High Viscosity.

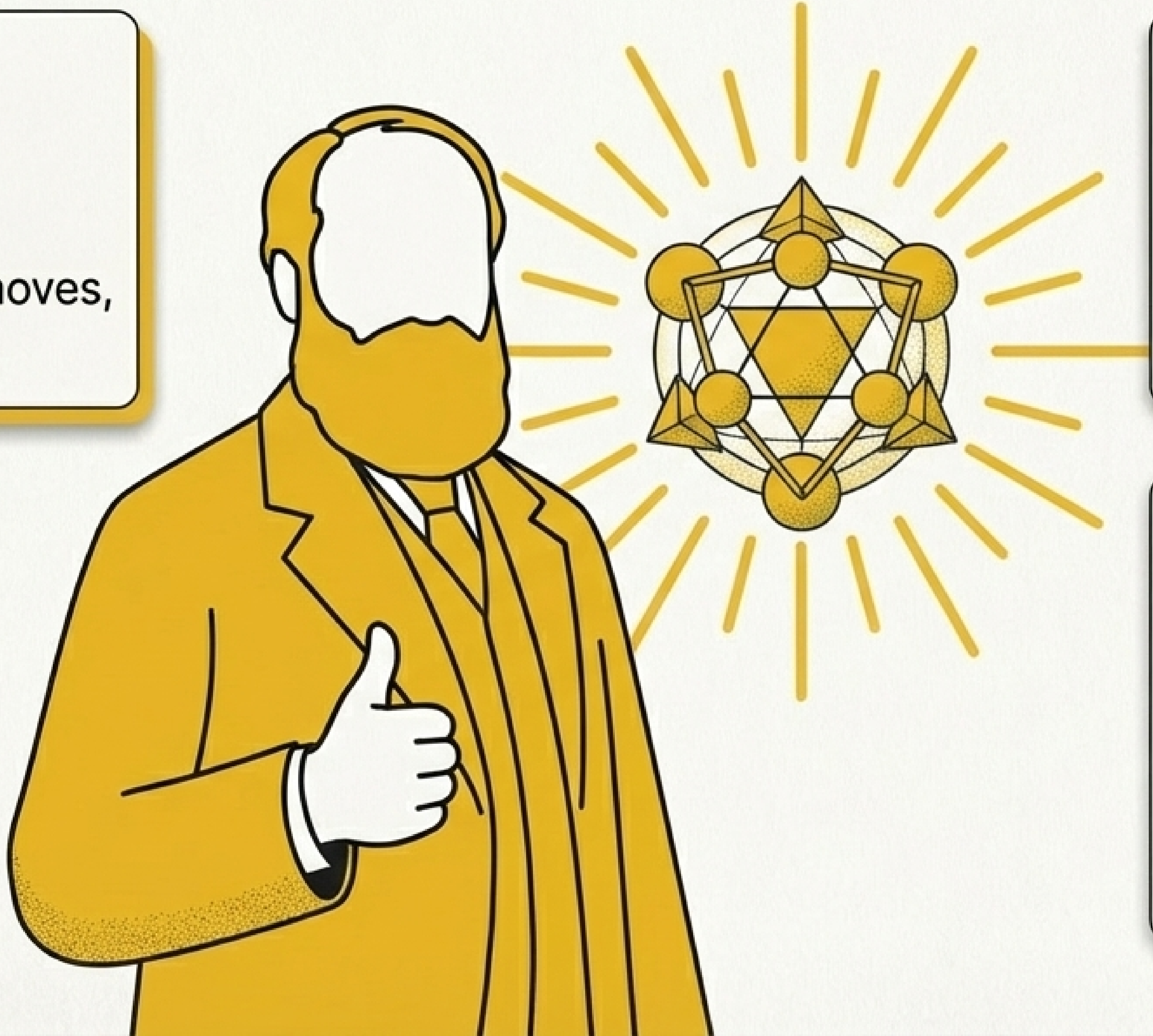
Juice = Low Viscosity.

Realm 2: What is Energy? | Was ist Energie?

Definition:

Energy is simply the ability to do work.

Without it, nothing moves, heats up, or lives.



The Unit:

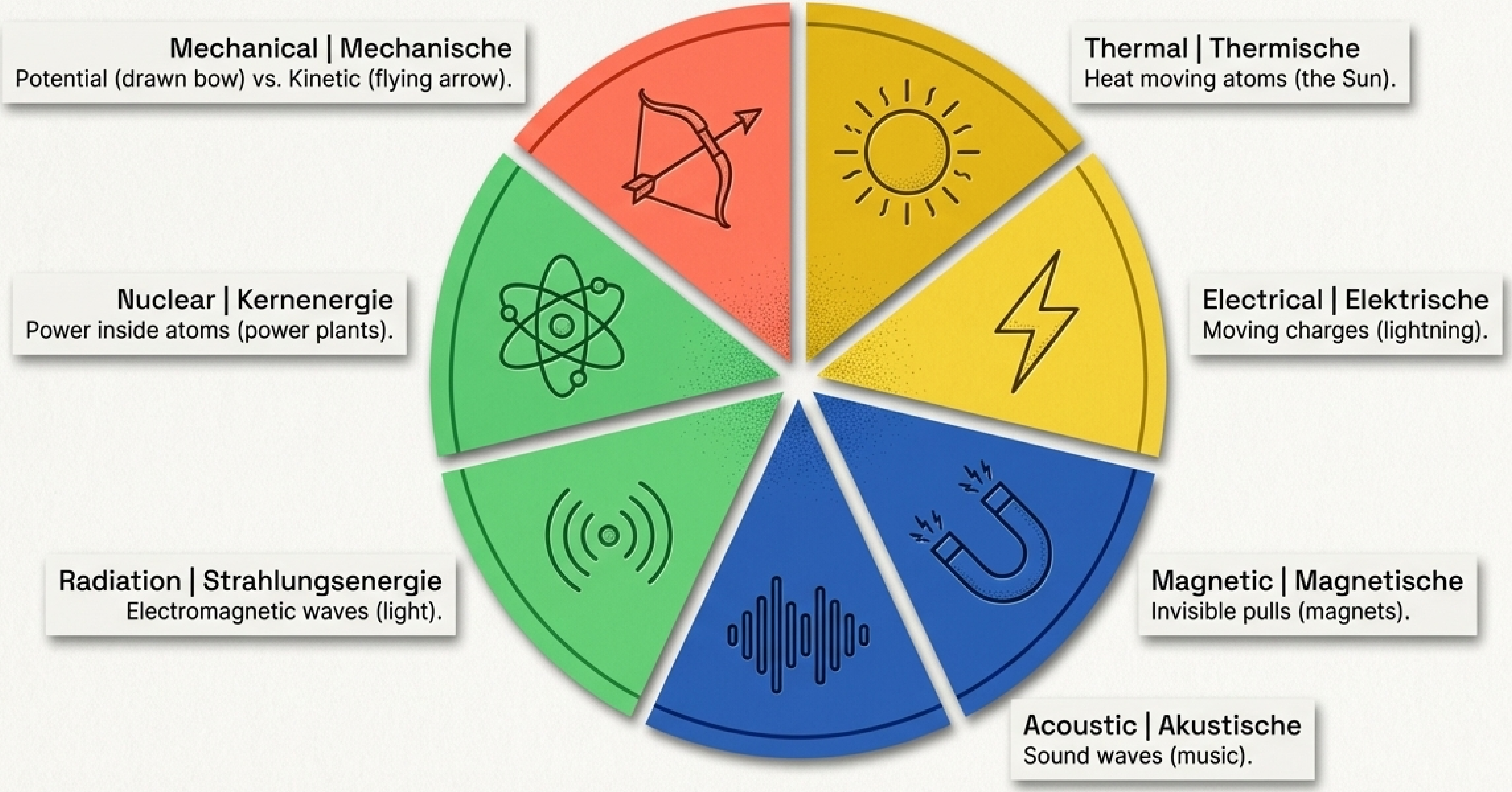
Measured in Joules (J).

(Fun fact: It takes 4190 Joules just to heat 1kg of water by 1°C!).

The Golden Rule:

Energy is never created or destroyed. It only transforms from one type to another (Energy Transformation | Energieumwandlung).

Forms of Energy | Energieformen



Realm 3: Forces | Kräfte

Forces are invisible pushes and pulls. You can't see them, but you can see what they do. Measured in Newtons (N).

Rule 1: They move things

Rule 2: They deform things

Force Multipliers | Kraftwandler

Pliers (levers) multiply our strength.

Excavators (hydraulics) multiply our strength.

The Ultimate Physics Cheat Code

$$F = m \times a$$

Force = Mass × Acceleration | Kraft = Masse × Beschleunigung

Want to make something speed up or change direction (**Acceleration**)? You need **Force**!

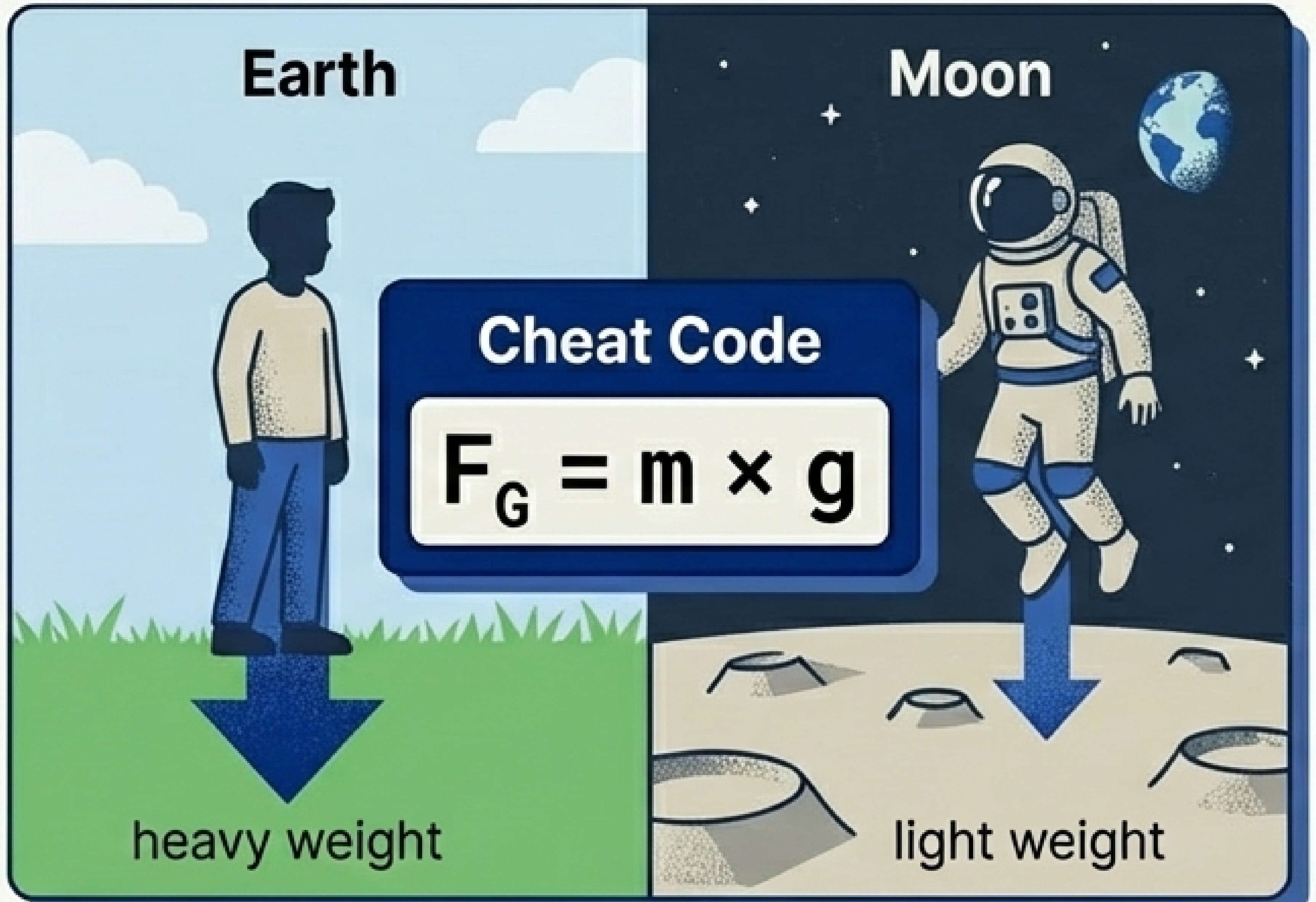
The heavier the object (**Mass**), the more Force you need. Pushing a bike is easy. Pushing a heavy truck requires massive **Force**.

Newton's Law of Inertia | Trägheitsgesetz:
Things like to stay exactly how they are until a force messes with them!

Mass vs. Weight | Masse vs. Gewichtskraft

Mass | Masse (kg)

How much ‘stuff’ you are made of. This **NEVER** changes. (60kg on Earth, 60kg on the Moon).



Weight Force | Gewichtskraft (N)

How hard gravity pulls on that stuff. This **CHANGES** based on where you are!

Cheat Code: $F_G = m \times g$

On Earth, gravity (g) is about 10 m/s². So a 60kg kid has a weight force of 600 Newtons!

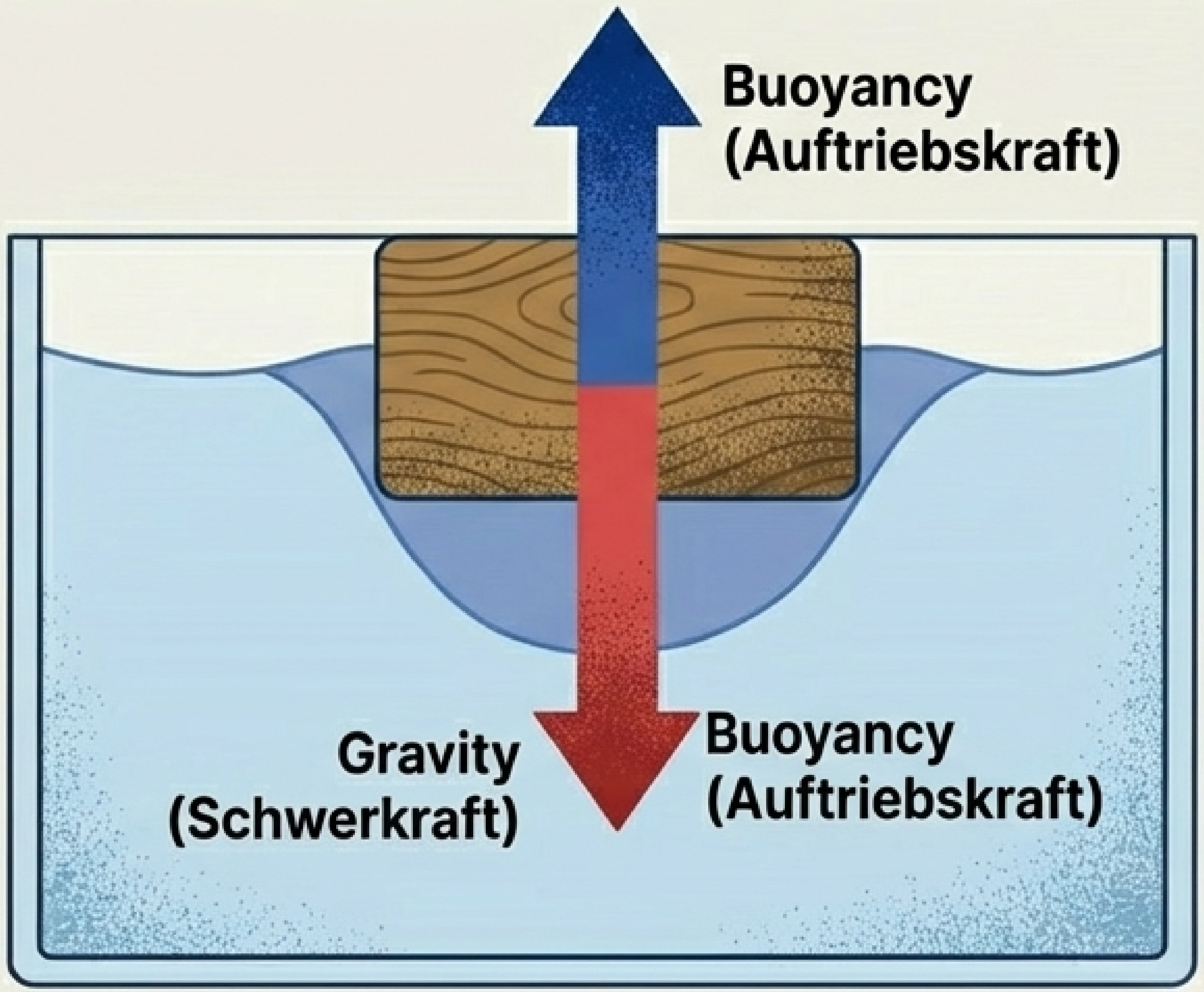
Buoyancy | Auftriebskraft

Why do heavy ships float? Because water pushes back!

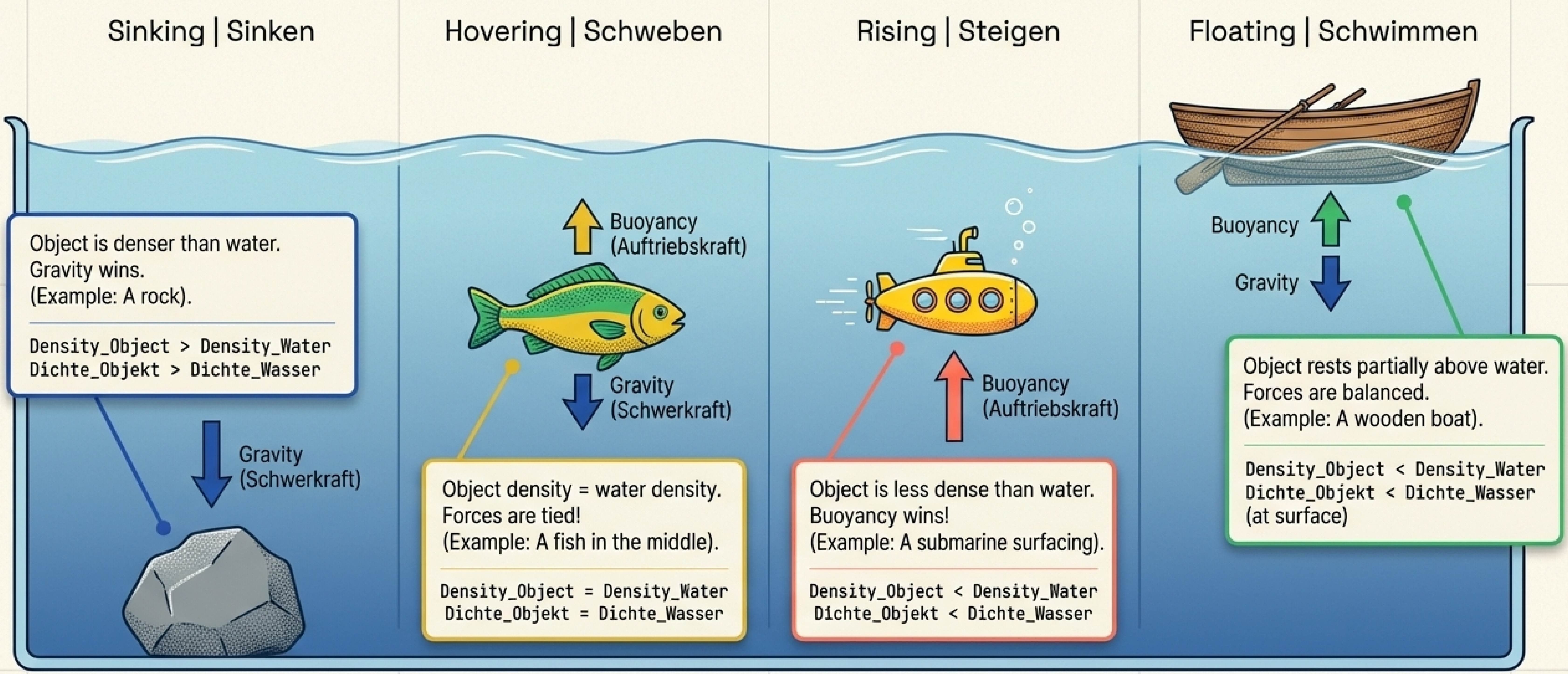
Archimedes' Principle:

The upward push (**Buoyancy**) is exactly equal to the **weight of the water** the object pushes out of the way.

The **denser the liquid, the stronger the push.** (This is why it's easier to float in salty ocean water than in a regular swimming pool!).



Sink, Float, or Hover? | Sinken, Schwimmen, Schweben?



NotebookLM

Realm 4: Biology | Biologie

Biology comes from the Greek *bíos* (life) and *lógos* (study). It's the science of living organisms!

The Bio-Squad:

1

Zoology | Zoologie

The study of animals (zóon).

2

Botany | Botanik

The study of plants (botaniké).

3

Microbiology | Mikrobiologie

The study of tiny, invisible life (mikrós).

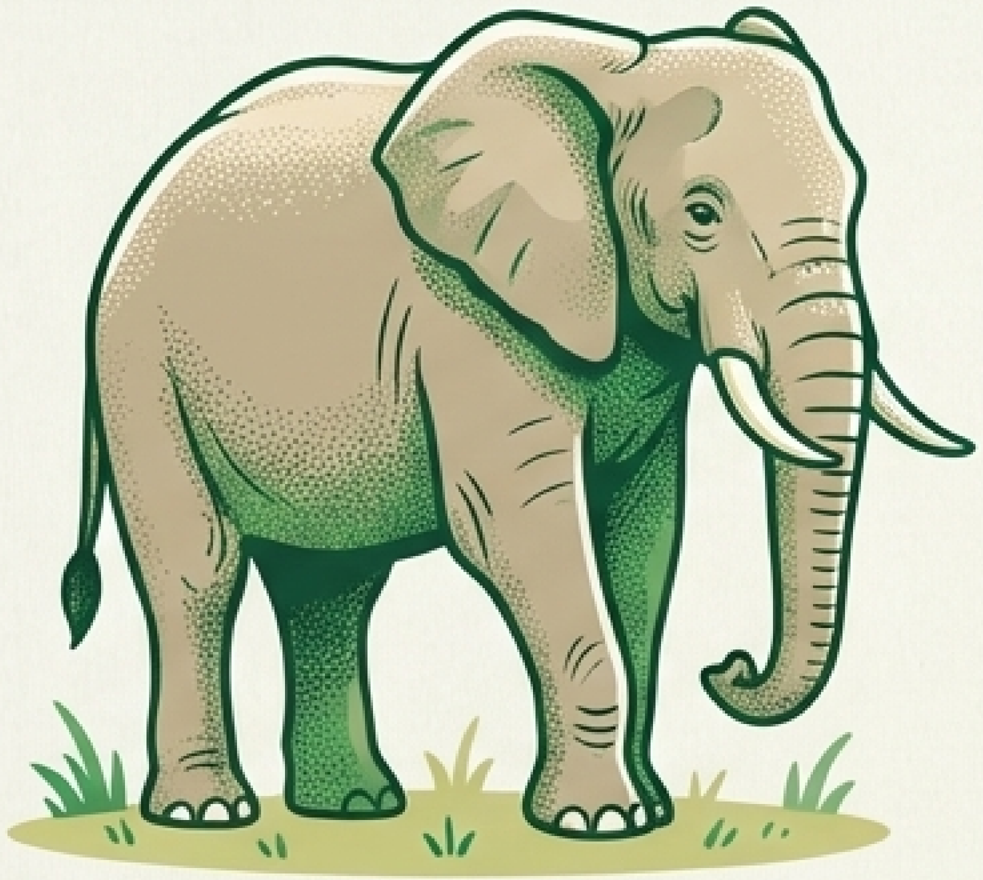
4

Ecology | Ökologie

How living things interact with their environment (oikos = household).

How to Spot a Living Thing | Kennzeichen des Lebens

To be considered “alive,” an organism must do ALL 6 of these:



1. Cells | Zellen:
Made of at least one cell.

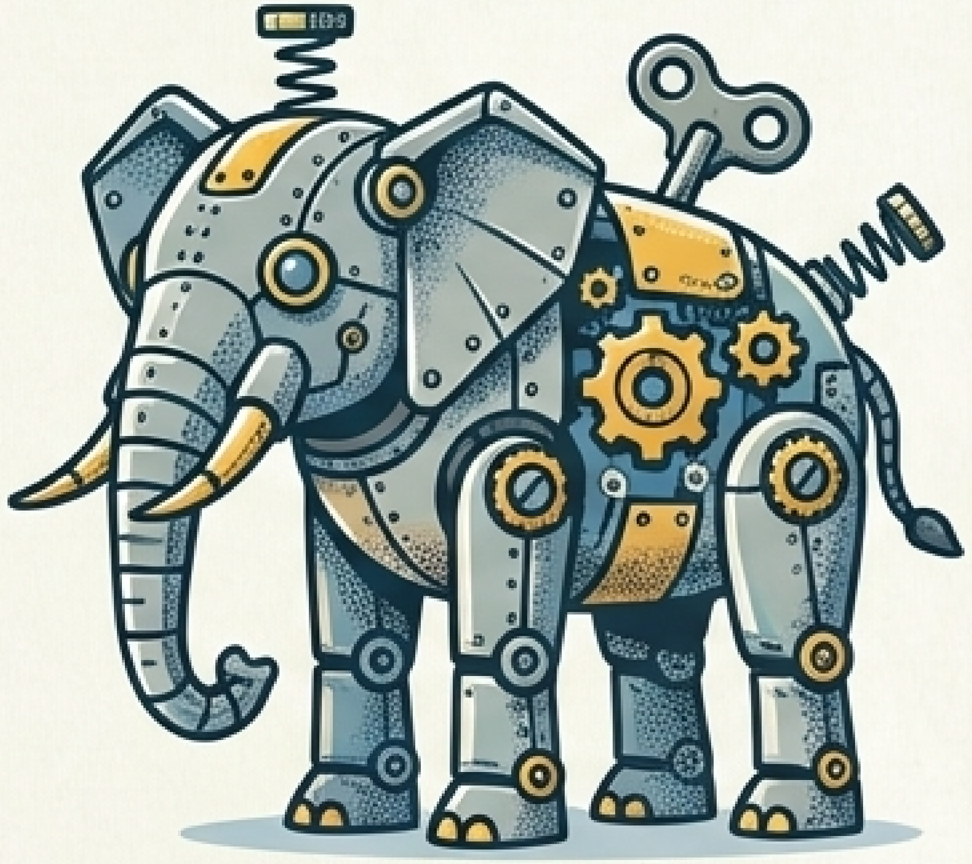
2. Metabolism | Stoffwechsel
Eats stuff, uses energy, makes waste.

3. Reacts to Stimuli | Reizbarkeit
Reacts to the world.

4. Reproduction | Fortpflanzung
Can create offspring.

5. Movement | Bewegung
Moves on its own.

6. Growth | Wachstum
Gets bigger.



Final Boss Checklist

Level Complete

Secret Language: Binomial Nomenclature | Binäre Nomenklatur

Every organism gets a 2-part Latin name: Genus + Species. (Example: *Homo sapiens*).

The Big 3 Formulas:

Density: $\rho = \frac{m}{V}$

Force: $F = m \times a$

Gravity: $F_G = m \times g$

You have unlocked all the knowledge. The VSN team wishes you massive success on the NW LZK on 09.03.2026. Go ace this!